



Colony PCR

Designing Life with AI

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Description

This method allows to check the length of the DNA in our cells to confirm whether the plasmid had been fully inserted.

Material

- Appropriate antibiotic (if needed for selection)
- Sterile MilliQ water
- 2x Taq Master Mix **or** Q5 High-Fidelity Master Mix
- Forward and reverse primers
- DNA ladder
- Loading dye
- Agarose powder
- 1x TBE buffer
- SYBR Safe DNA stain (or equivalent safe DNA dye)

Steps

I – Primers design for Colony PCR

- Target the **junctions of the assembled sequence**.
- Avoid sequencing unreliable regions: first 100 bp and after ~ 1 kb with Sanger sequencing are often noisy.
 - Place primers ~ 100 bp before the junction.
- Use Primer3 (<https://primer3.ut.ee>) to make the sequence:
 - “Pick left primer” = forward.
 - “Pick right primer” = reverse.
 - If you have the possibility, put 2 for “GC clamp”.
- Copy the suggested sequence into Benchling and check for:
 - Its position / match on the new assembled plasmid,
 - Acceptable secondary structure.

II – Colony preparation

- Pick single colonies.
- Streak them on a petri dish (with the corresponding antibiotic if necessary) and put the petri dish in a static incubator.
- Inoculate them in a PCR tube containing 30 μ L of water.

III – PCR setup

PCR mastermix recipe:

Ingredients	Volumes [μL]
2x Taq or Q5 HF Master Mix	25
Forward primer	2 / 2.5
Reverse primer	2 / 2.5
DNA template	1
MilliQ water	19–21
Total	to 50

Run PCR with fixed settings except for:

- Annealing temperature depending on the primers used.
- Extension time adjusted to amplicon size.

IV – Gel verification

- Prepare agarose gel (see Agarose gel preparation protocol).
 - Weigh 0.5 gr of agarose (to get to 1%)
 - Mix with 50 mL of 1x TBE buffer
 - Heat it 1 minute and 30 seconds in the microwave
 - Let it cool a bit
 - Add 1 μL of Sybrsafe (1 μL per 50 ml gel) and mix gently
 - Pour the gel in a rack with the appropriate comb
 - Let it solidify for around 20 min
- Load 5 μL of DNA ladder and PCR product + add 1 μL of loading dye per sample if necessary.
- Run gel at 135 V for 15 min.
- Visualize DNA bands under UV light.

Manufacturer's instructions depending on the master mix used

For **2x Taq Master Mix** from Vazyme:

<https://www.vazymeglobal.com/product-center/conventional-pcr/2-taq-master-mix>

For **Q5 High-Fidelity 2x Master Mix** from NEB:

<https://www.neb.com/en/products/m0492-q5-high-fidelity-2x-master-mix>

Thermocycling conditions for a routine PCR (Q5): initial denaturation 98 °C for 30 s; 25–35 cycles of 98 °C for 5–10 s, 50–72 °C for 10–30 s, and 72 °C for 20–30 s/kb; final extension 72 °C for 2 min; hold at 4–10 °C. Use of the NEB T_m Calculator is highly recommended.